[DISTRIBUTIONAL]

Towards Automated Evaluation of LLM Applications

Erin LeDell, Ph.D.
Chief Scientist, Distributional
September 9, 2025
AutoML 25
New York, NY, USA

[DISTRIBUTIONAL]

01 Introduction
02 LLM & AI Agent Applications
03 Eval Fundamentals
04 Eval in Dev vs. Prod
05 LLM Analytics & Testing
06 Towards Auto-Eval

[DISTRIBUTIONAL]

01 Introduction

Hello! 👋

Chief Scientist at Distributional, Inc. (USA)
Building an automated LLM eval, analytics and testing platform (DBNL)

AutoML history:

- Working in AutoML since 2015 (LLMs since 2022)
  - Creator of H2O AutoML (2017), as Chief ML Scientist at H2O.ai (2015-2023)
  - Co-creator of AutoML Benchmark (AMLB) (2019)
  - Long history of eval & benchmarking ML (back to 2013)
- Scalable stacked ensemble work from my PhD (2015) became H2O AutoML
- Open source software at: https://github.com/ledell

02 LLM & AI Agent Applications

LLM Applications

Common design patterns for LLM applications:

- **Q&A Chatbot** (e.g. ChatGPT w/ optional custom prompt, or a fine-tuned model)

- **Retrieval Augmented Generation (RAG)**
  - Augment an LLM with a knowledge base (document corpus or db) - combines a retriever with a generator.
  - Uses information retrieval techniques such as document embeddings to surface relevant context to ground the model's response in external knowledge.

- **Structured output**
  - Data extraction from text
  - Optionally interface with APIs
  - Response in JSON, CSV, Markdown, etc.

Unique Challenges of LLM Applications

Evaluation has always been multi-dimensional — but LLM applications blow open the space of dimensions, making it more subjective, context-dependent, and harder to automate.

Challenging aspects of LLMs:

- **Probabilistic outputs**: No single “correct” answer; evaluation depends on context.
- **Dynamic models & APIs**: Frequent model updates of commercial models.
- **Context dependent:** Quality of the response depends on the domain of the question, quality of the input and user intent.
- **Subjective metrics:** Helpfulness, tone, bias, safety are hard to quantify.
- **Evaluation cost and scalability:** Human labeling and expert review do not scale and cannot easily be automated (unless you resort to LLM-as-Judge).

Beware: Silent Model Updates 🤫

Even if you are using the same exact model number, OpenAI and other model providers have a history of updating models without releasing a new version, causing silent shifts.

Model	Behavior Change
`gpt-4o-2024-08-06`	Output inconsistency — sometimes repeated or missing items in lists
Default `gpt-4o` (Oct 2024)	Regression in function calls (returned as plain text, not structured), worse adherence to instructions !!
`gpt-4-turbo` (default, mid-2024)	Quality drop: more formatting errors, failures to follow exemplars in JSON or structured outputs
`gpt-4-1106-preview` → same label, early 2024	“Laziness” bug (skipping tasks, incomplete answers), fixed silently
`gpt-3.5-turbo` (Mar 2023 vs Jun 2023)	Performance drift: prime/composite accuracy dropped (84% → 51%), code formatting degraded, safety refusals increased

Beware: Silent Model Updates 🤫

Hot off the presses: Today, Anthropic released a statement about two model degradations that were reported last week. 🔥

To the right, commentary from a podcast about the surprising “lack of evals” at the top LLM companies. 🔥🔥🔥

status.anthropic.com

Model output quality

Incident Report for Anthropic

SUBSCRIBE TO UPDATES

Investigating

Last week, we opened an incident to investigate degraded quality in some Claude model responses. We found two separate issues that we've now resolved. We are continuing to monitor for any ongoing quality issues, including reports of degradation for Claude Opus 4.1.

Resolved issue 1 - A small percentage of Claude Sonnet 4 requests experienced degraded output quality due to a bug from Aug 5-Sep 4, with the impact increasing from Aug 29-Sep 4. A fix has been rolled out and this incident has been resolved.

Resolved issue 2 - A separate bug affected output quality for some Claude Haiku 3.5 and Claude Sonnet 4 requests from Aug 26-Sep 5. A fix has been rolled out and this incident has been resolved.

Importantly, we never intentionally degrade model quality as a result of demand or other factors, and the issues mentioned above stem from unrelated bugs.

We're grateful to the detailed community reports that helped us identify and isolate these bugs. We're continuing to investigate and will share an update by the end of the week.

Posted 14 hours ago. Sep 09, 2025 - 00:15 UTC

This incident affects: claude.ai, console.anthropic.com, api.anthropic.com, and Claude Code.

swyx @swyx

Claude Code: no evals

[well known code agent company]: no evals

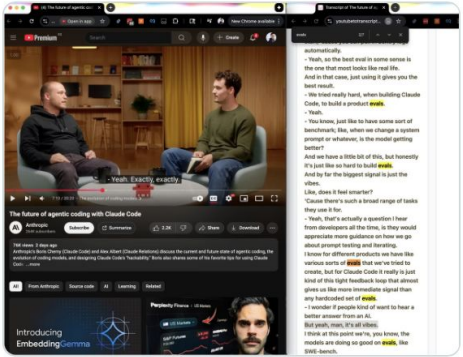
[well known code agent company 2]: kinda halfassed evals

[leading vibe coding company]: no evals

[ceo of company selling you evals]: mmmmm yess all my top customers do evals, you should do evals

[vc's in love with ceo of evals company]: mmmmm yes all my top founders do evals, must do evals

(NOTE: i -do- also think that evals are impt, but the eval pilled ai engineers have also noticed that it is not a strict requirement for success and, at least for 0-to-1 stage, may even be anticorrelated, think thru why)



Lenny Rachitsky @lennysan · Sep 4

Trend I'm following: evals becoming a must-have skill for product builders and AI companies.

It's the first new hard skill in a long time that PMs/engineers/founders have had to learn to be successful. Th...
Show more

3:08 PM · Sep 4, 2025 · **464.6K** Views

120 146 1.2K 1K

Traditional ML vs. LLM Applications

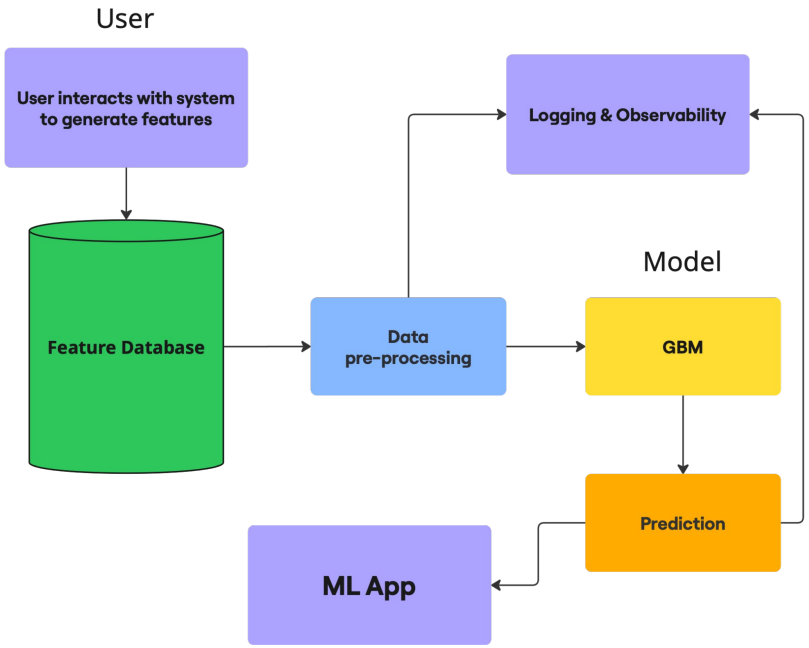
	Traditional ML	LLM Applications
Outputs	Discrete, structured	Open-ended, probabilistic
Metrics	Accuracy, AUC, RMSE, inference speed, …	Correctness, safety, helpfulness, tone, bias, response time, …
Evaluation Cost	Low (fully automated)	High (can be semi-automated; typically there's humans in the loop)
Stability	Fixed models/test sets, however live data can shift over time	Models & APIs drift frequently, input data can also shift over time

ML Applications

Example **ML pipeline:**

- This is a prototypical ML pipeline where features are generated directly by the user and/or by some system.
- After (optional) data-preprocessing steps, those features are given to the model, which makes predictions.
- The predictions are then used some downstream in the application (e.g. to determine which product to show in an e-commerce website).

Machine Learning Production System



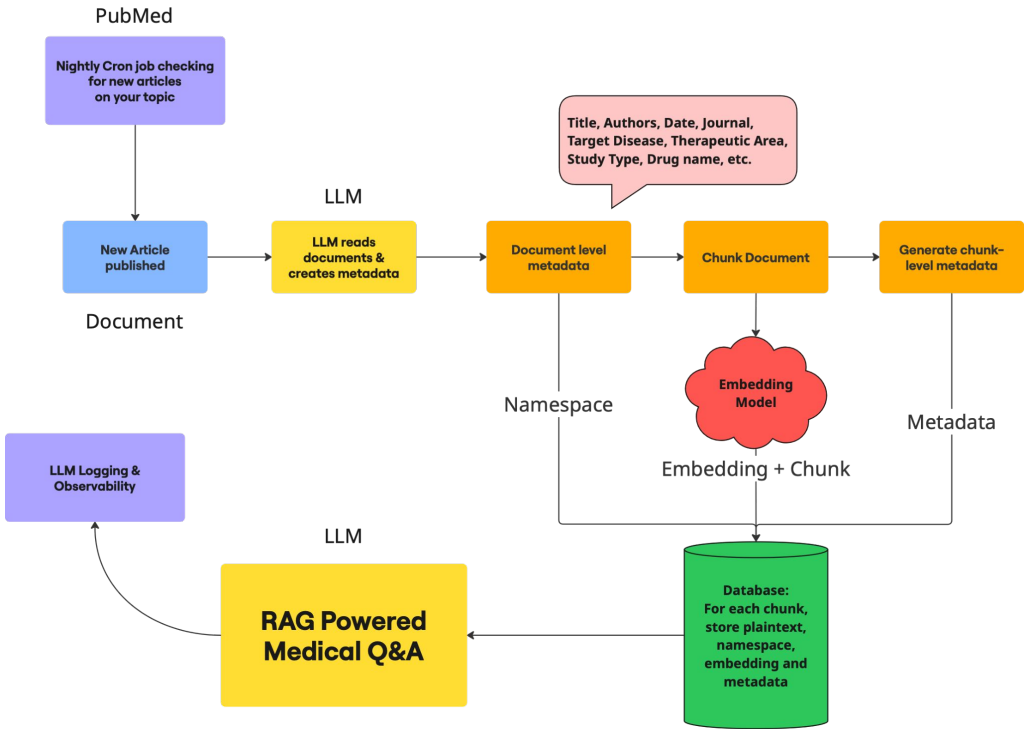
LLM Applications

- Modern AI Applications are **complex** and increasingly **use Generative AI.**
- **LLMs** are a common feature across various components of AI applications including even data pipelines (ETL).

Example **medical research LLM pipeline**:

- Application which uses an LLM to processes academic articles in service of regularly updating a RAG corpus.
- Derives structured data representations from the documents and metadata tags.
- Need to evaluate the LLM output in the pipeline, as well as the main RAG application (eval both retrieval and generation metrics)

Data processing new RAG corpus documents



Properties of LLM Systems

Modern AI systems typically contain LLMs and thus these systems have a number of interesting and more complex properties when it comes to eval and testing:

- **Multi-component:** LLM apps depend on pipelines, databases, and APIs, so changes in one component can cause unexpected effects elsewhere.
- **Non-stationary:** Behavior shifts over time as data, language use, or underlying models change, leading to regressions without code changes.
- **Non-deterministic:** The same input can yield different but valid outputs, making exact-match tests unreliable.

Properties of AI Agents 🤖

AI Agents are LLM-based systems that act as a controller that interprets a user request and calls APIs, tools, or code execution environments:

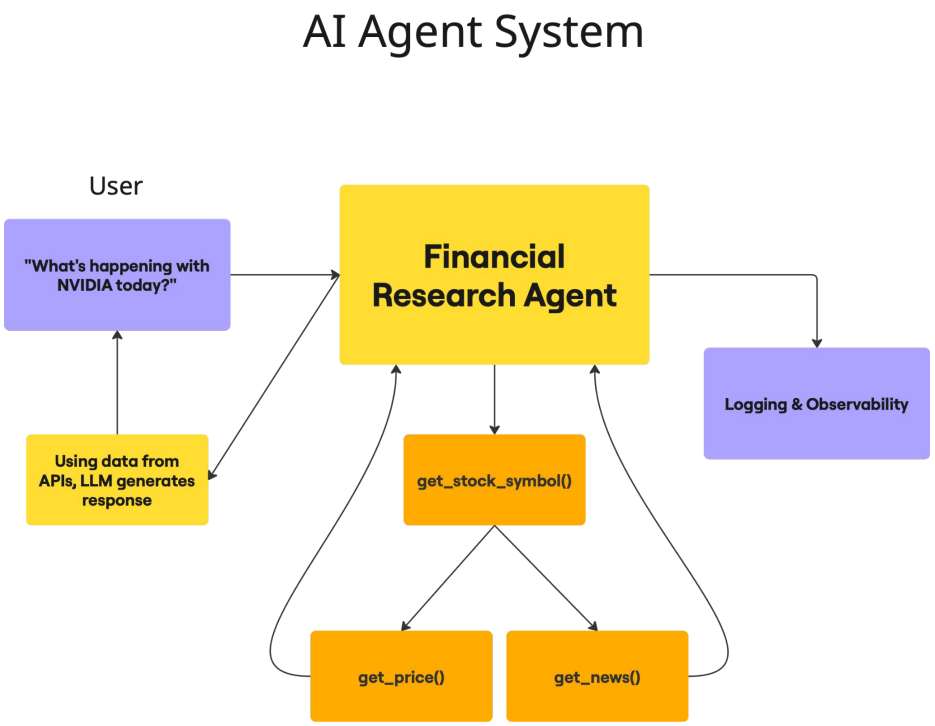
- **Interactive & Environment-Dependent:** Agents act through sequences of tool calls and API interactions. Their behavior depends not just on the model but also on changing external systems.
- **Stochastic & Path-Sensitive:** The same task may yield different action sequences. Small errors in intermediate steps can cascade, leading to very different outcomes.
- **Long-Horizon & Outcome-Oriented:** Success depends on coordinating multiple steps. Evaluation must focus on whether the agent completes the overall task, not just local accuracy.

AI Agent Systems

- Modern AI Applications are more commonly **“agentic”**, meaning they have **access to external tools and can take actions.**
- Due to standard protocols like MCP, building agents is becoming more common.

Example **Financial Research Agent**:

- An LLM chat interface with access to several API-backed tools that can retrieve live and historical financial data and news about companies.
- Automatic Tool Orchestration: AI decides which tools to use and how to combine results for the user.



03 Eval Fundamentals

LLM Eval: 📊📈🔬

LLMs have a rapidly developing ecosystem of Eval metrics, methodologies and frameworks. However, these metrics are <u>less universally understood and more subjective</u>.

While evaluation metrics for LLMs are different than traditional ML, there are some **overlapping and analogous goals**:

- **Accuracy:** Does it give the correct answer?
- **Factuality:** Are its claims true?
- **Helpfulness:** Is it useful to the user?
- **Safety:** Does it avoid harmful or biased outputs?
- **Generalization:** Can it handle tasks it wasn't explicitly trained on?

Differences:

- The evaluation metrics themselves look very different than in ML. The main difference is that they are functions that take **text as input**. e.g. BLEU, ROUGE, LLM-as-Judge

LLM Eval: Automatic Text Eval Metrics

The most basic level of LLM Eval includes computing “automatic” text-related metrics.

Limitations:

- Most require a ground truth response to evaluate (requires human to generate these responses).
- These are not very useful, unless you have a very simple task such as a summarization app.
- To make use of these, you can track them over time and look for outliers in the scores, which could be indicative of a bug or error in your system that warrants further manual investigation.

Metric	Eval Focus
BLEU	N-gram overlap; widely used in machine translation.
ROUGE	Recall-based overlap; standard for summarization.
Perplexity	Measures fluency/naturalness.
BERTScore	Embedding similarity; robust for paraphrasing and summarization/QA.
Accuracy/F1	Classification performance; sentiment, intent detection.

LLM Eval: Humans in the Loop

Humans (and their opinions) are often involved in the LLM evaluation process, which is very different from how automated evaluations of ML systems works.

Humans create **Golden (validation) datasets**:

- They write input for these datasets to mimic real life input
- They also can write correct/good responses or edit LLM-generated responses
- They can also provide binary/categorical (👍vs 👎) feedback on responses
- You can annotate traces with opinionated text or structured data (for error analysis)

Limitations:

- Golden datasets evaluated in development might not reflect the true distributions of the input data (how users interact with your system).
- Golden datasets may only reflect certain conditions that you anticipated where as real life usage may have many wild deviations from that.

LLM Eval: LLM-as-Judge

What is LLM-as-Judge? ⚖️

- **Definition:** Using an LLM to evaluate the quality, accuracy, or usefulness of responses generated by LLM models or applications.
- Often used to **replace or supplement human annotators** in large-scale evaluations.
- **Fully automated**, does not require ground truth labels and because of these two attributes, it is widely used.
- Likert scores (1-5) or Binary (0/1) scores are commonly used.

Limitations:

- **Bias:** LLMs may favor stylistic features, verbosity or similar outputs.
- **Prompt sensitivity:** Evaluation quality can vary with prompt design.
- **Validation required:** LLM judgments often need correlation with human evaluations to be trusted – evaluating the judge should not be automated.

More info: https://hamel.dev/blog/posts/llm-judge/

LLM Eval: LLM-as-Judge

```
BINARY JUDGE — FINANCIAL RESEARCH AGENT (3 CHECKS)

Inputs:
- Question
- Agent Answer
- (Optional) Trusted Snapshot (quotes/news)

Checks (True/False):
1) Accuracy & Freshness — Facts/figures have timestamp + source; if snapshot provided, numbers/news
   align within basic tolerance/time window.
2) Reasoning & Relevance — Uses data to answer the question directly; no unwarranted speculation.
3) Clarity — Clear, concise, professional.

Decision:
- overall_pass = True only if all three checks are True.

Output (JSON):
{
  "accuracy_freshness": true | false,
  "reasoning_relevance": true | false,
  "clarity": true | false,
  "overall_pass": true | false,
  "notes": "1 short sentence on why."
}
```

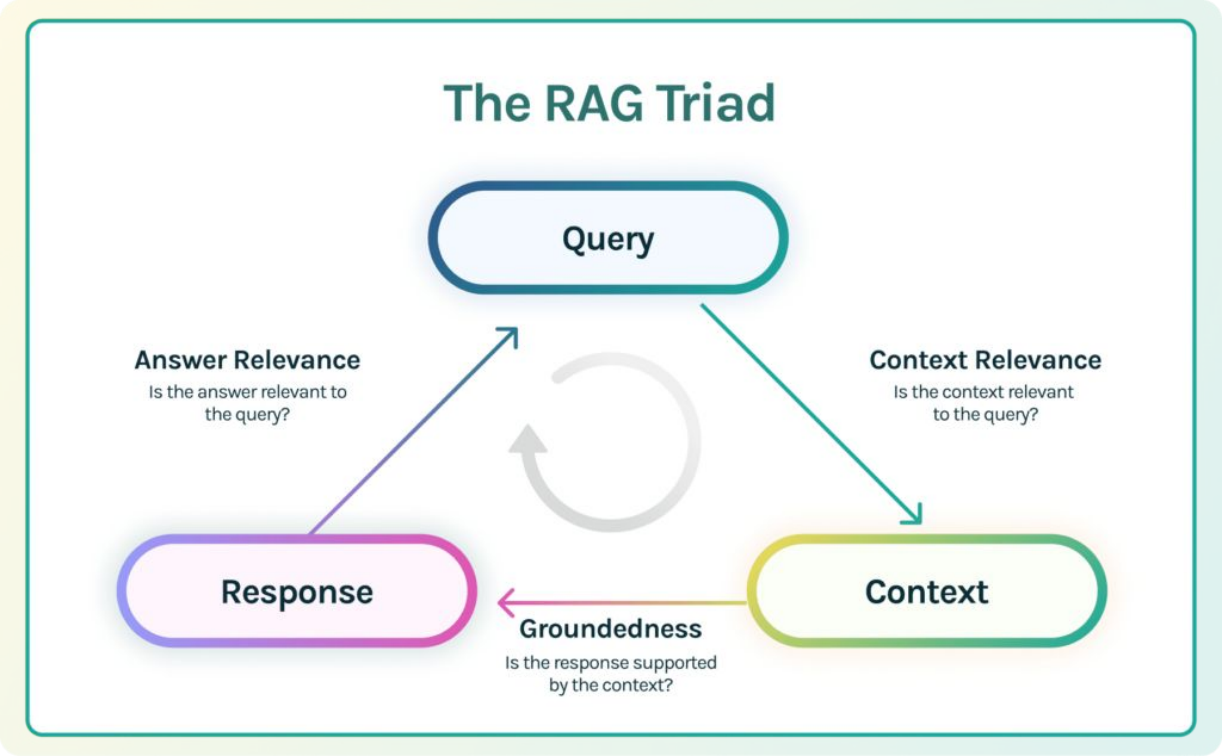
Binary responses in Judges are considered best practice over Likert scales (1-5) even though that is very common.

More information here: https://tinyurl.com/hamel-likert

LLM Eval: LLM-as-Judge

```
SIMPLE DATA VALIDATION JUDGE — FINANCIAL RESEARCH AGENT

Inputs:
- Question, Agent Answer
- Trusted Snapshot: quotes[{sym, price, hi, lo, cur, t}], news[{headline, pub, t}]

Checks:
1) Price claims → must include {sym, value, t}.
   PASS if currency matches AND (|value−price|/price ≤ 0.5% OR hi ≤ value ≤ lo).
2) News claims → must include {headline/event, pub, t}.
   PASS if headline+publisher match a snapshot item within ±5 min.
3) Freshness → if the question implies “today/now”, require t is same day (≤15 min ok).
4) Missing timestamp/source → FAIL that claim.

Decision:
- PASS only if all claims pass; otherwise FAIL with reasons.

Output (JSON):
{
  "pass": true|false,
  "failures":[{"type":"price|news","id":"AAPL|headline","reason":"delta>0.5%|no-ts|stale|currency"}]
}
```

LLM Eval: RAG Metrics



Context Relevance

- Ground truth not needed

Groundedness

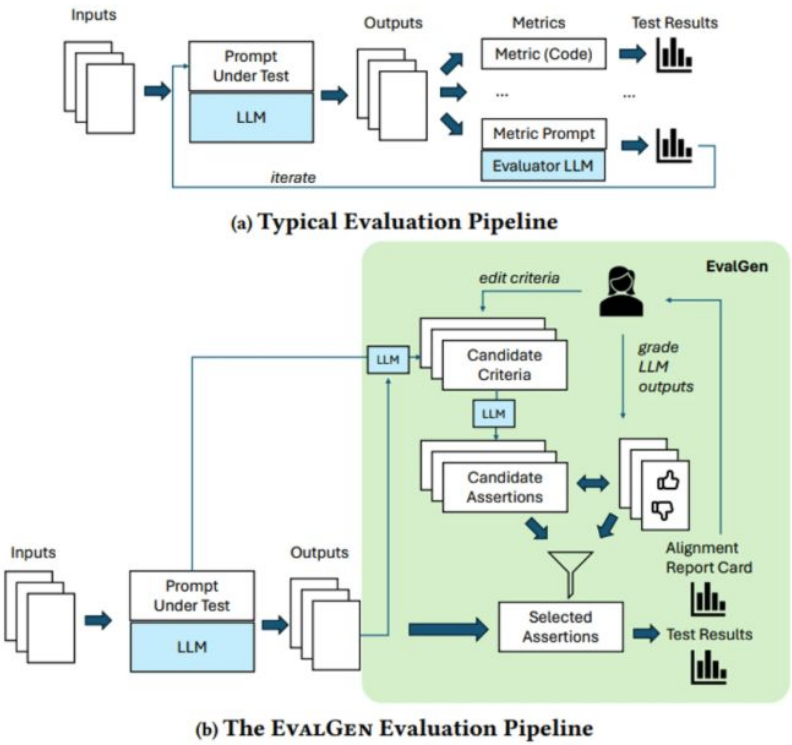
- Ground truth answers needed or LLM-as-Judge

Answer Relevance

- Ground truth answers needed or LLM-as-Judge

LLM Eval: EvalGen for Judge Alignment

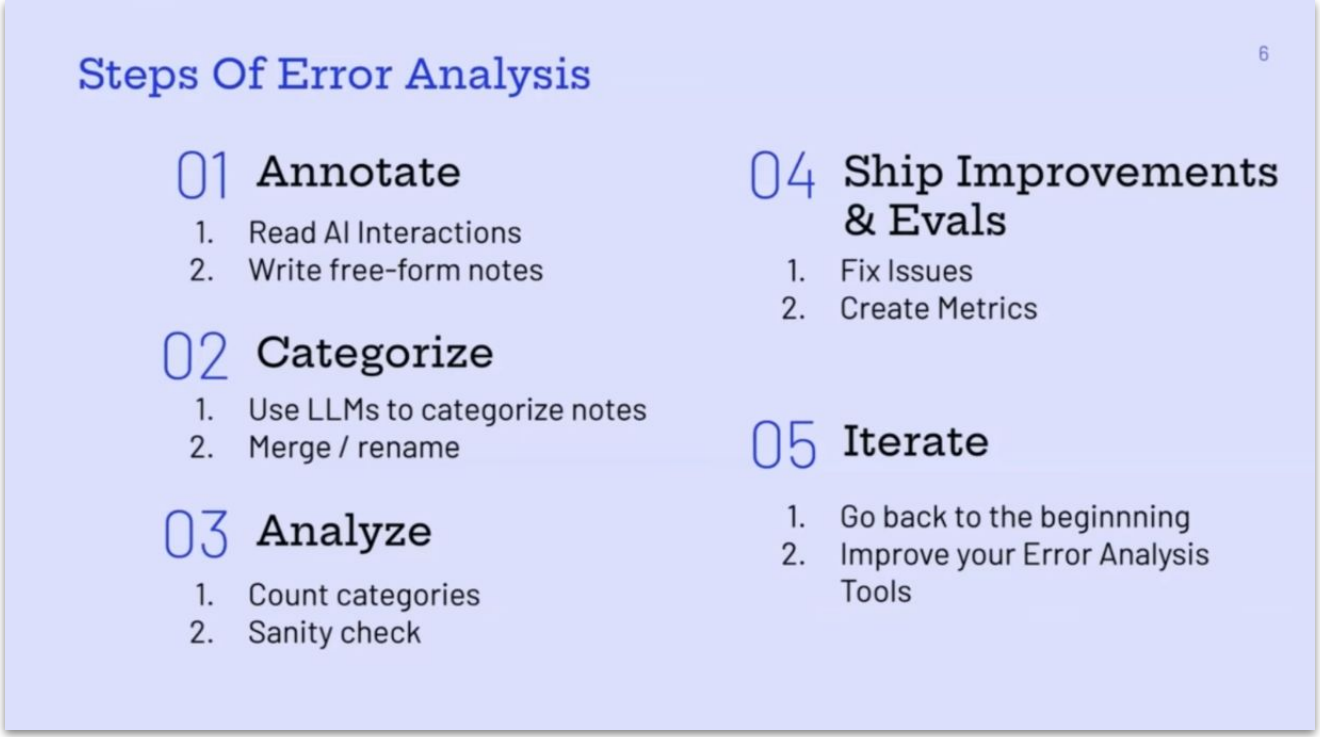
- **EvalGen** is a framework for automating the evaluation of language model outputs using LLMs-as-judges.
- Helps align your evaluation data to human feedback (alignment).
- While generating candidate implementations (Python functions, LLM grader prompts), EvalGen asks humans to grade a subset of LLM outputs; **this feedback is used to select implementations that better align with user grades.**



https://arxiv.org/abs/2404.12272

LLM Eval: Error Analysis

LLM error analysis is the process of systematically examining the mistakes made by a large language model (LLM) in order to understand what kinds of errors occur, why they happen, and how to reduce them.



Source: https://www.youtube.com/watch?v=qH1dZ8JLLdU

04 Eval in Dev vs. Prod

Dev vs. Prod: Offline vs. Online Eval

Offline Evaluation:

- Eval that occurs in **development** or **deployment** environments.
- In service of model selection / model upgrading for an ML application.
- Includes benchmarking

Online Evaluation:

- Eval that occurs in **production** environments.
- In service of model monitoring, drift detection.

Both can include:

- Evaluation with off-the-shelf model perf based metrics & also business metrics.
- Evaluate model fairness and bias.
- Evaluation of latency / inference speed.
- Evaluation of model memory / compute requirements.

Development: Offline Eval 🛠️🧪

Evaluation in the **development process** of AI systems is called Offline Eval:

Core Properties

- **Golden Datasets**: curated, high-quality labeled examples for the target task
- **Repeatable & Reproducible**: same prompts always produce comparable scores
- **Controlled Environment**: evaluation without live user noise or feedback loops
- **Low Cost & Safe**: no risk of harming user experience during testing

Prompt & Dataset Testing

- Use representative **prompt–response pairs** for domain tasks
- Evaluate consistency across paraphrased or reformulated prompts

Development: Offline Eval 🛠️🧪

Offline Metrics

- Text quality: BLEU, ROUGE, BERTScore
- Semantic alignment: embedding similarity, entailment scores
- Task-specific: exact match (QA), accuracy (classification), pass@k (code)

Human & Synthetic Judging

- Humans evaluate **correctness, clarity, safety**
- “LLM-as-Judge” if you increased automation, scalability and lower cost

Robustness & Safety Checks

- Adversarial prompts, jailbreak attempts
- Bias and toxicity screening
- Stress testing with noisy or incomplete inputs

Development: Open Source Eval Metrics Libraries

There's a number of **off-the-shelf open source Eval packages** available in Python (used primarily for offline eval):

- **Inspect:** Python library for LLM evals by J.J. Allaire
- **evals:** By OpenAI
- **DeepEval:** Unit testing framework for LLM outputs
- **RAGAS:** Specific to RAG metrics
- **TruLens:** Also focuses on RAG metrics, has a UI

```python
from ragas import evaluate
from datasets import Dataset
import os

os.environ["OPENAI_API_KEY"] = "your-openai-key"

# prepare your huggingface dataset in the format
# Dataset({
#     features: ['question', 'contexts', 'answer', 'ground_truths'],
#     num_rows: 25
# })

dataset: Dataset

results = evaluate(dataset)
```

Production: Online Eval 🚀🌐⚙️

Evaluation **in production** LLM systems is called Online Eval:

Monitoring & Metrics

- Track prediction quality and LLM output consistency over time
- Detect data drift using statistical tests (e.g., KS test, Chi-squared)
- Trigger alerts when performance drops below thresholds

Experimental Techniques

- A/B Testing: Compare multiple models by exposing them to different user groups
- Shadow Testing: Run candidate models alongside the live system without user impact

Feedback Loops & Risks

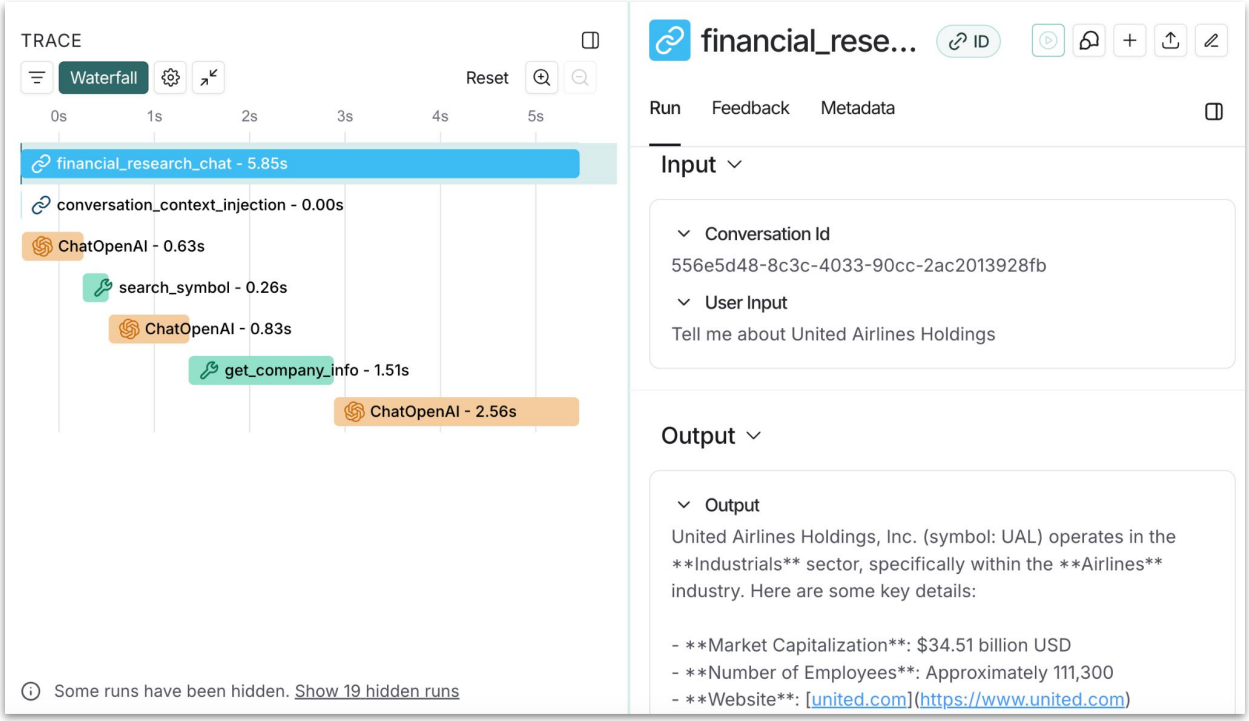
- Be aware of feedback loops, where model outputs influence future data
- Important in recommender systems, ads, finance, and adaptive LLM applications

Online Eval: LLM Logging Systems

How do we generate or collect LLM trace data for evaluation?

You need some way of **logging your traces** in both development and production settings.

There's a number of open source or free logging systems. (e.g. LangFuse, LangSmith, etc)



[DISTRIBUTIONAL]

05 LLM Analytics & Testing

LLM Analytics

Since LLM Applications are text based, we can perform a variety of of **text analytics** on the inputs/outputs and full traces. We can make use of well-known data science methods like clustering, text embeddings, anomaly detection.

Topic clustering

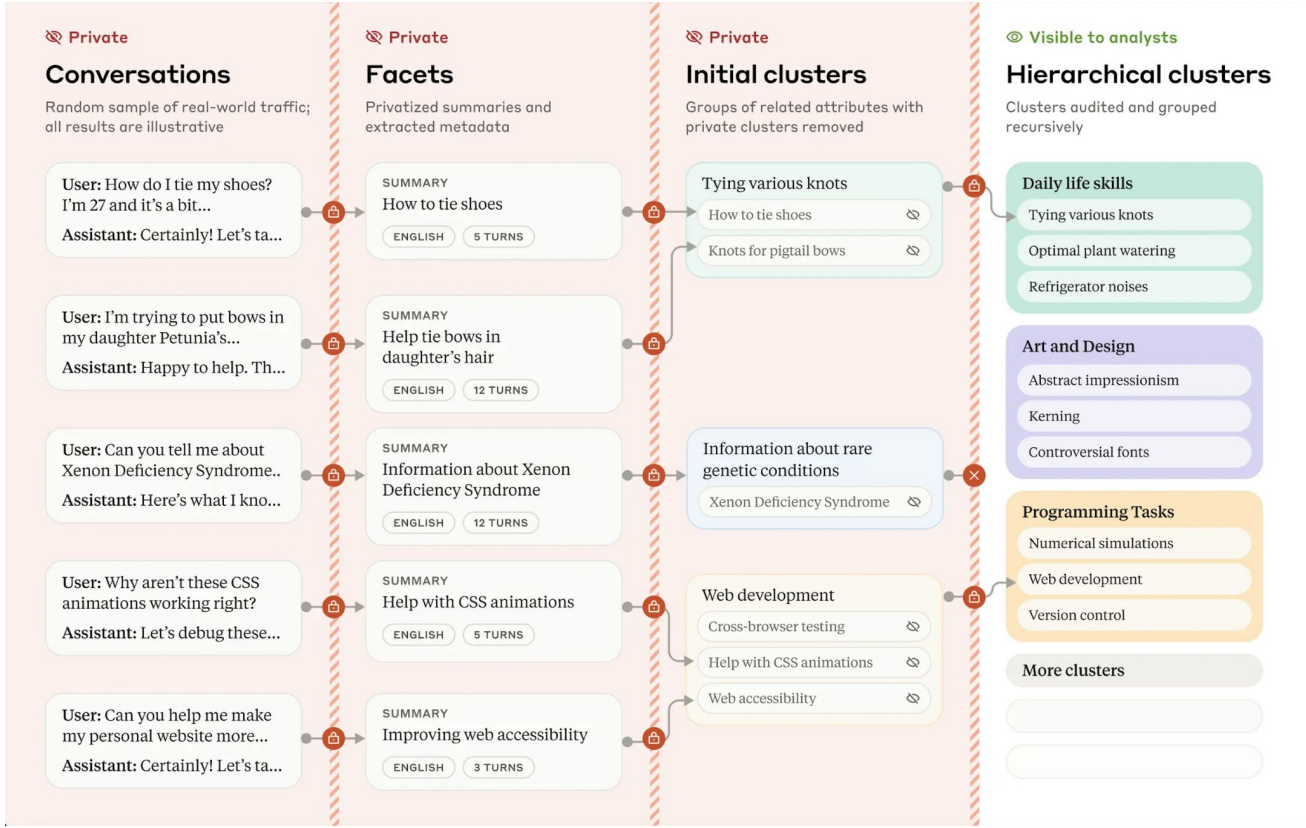
- **Clio** (Claude Insights and Observations) is a Dec 2024 paper by Anthropic about how to do privacy preserving clustering, in an attempt to understand how people are using Claude (e.g. what topics being discussed, what language, etc).
- Refer to Jason Liu's Kura for an open source reproduction of Clio: https://github.com/567-labs/kura
- Distributional (DBNL) also has an implementation of this (to be released very soon!)

Anomaly detection

- You can do anomaly detection on live production systems using standard anomaly detection methods on derived features (response_time_ms, total_token_count) or use the input embeddings as features as well.

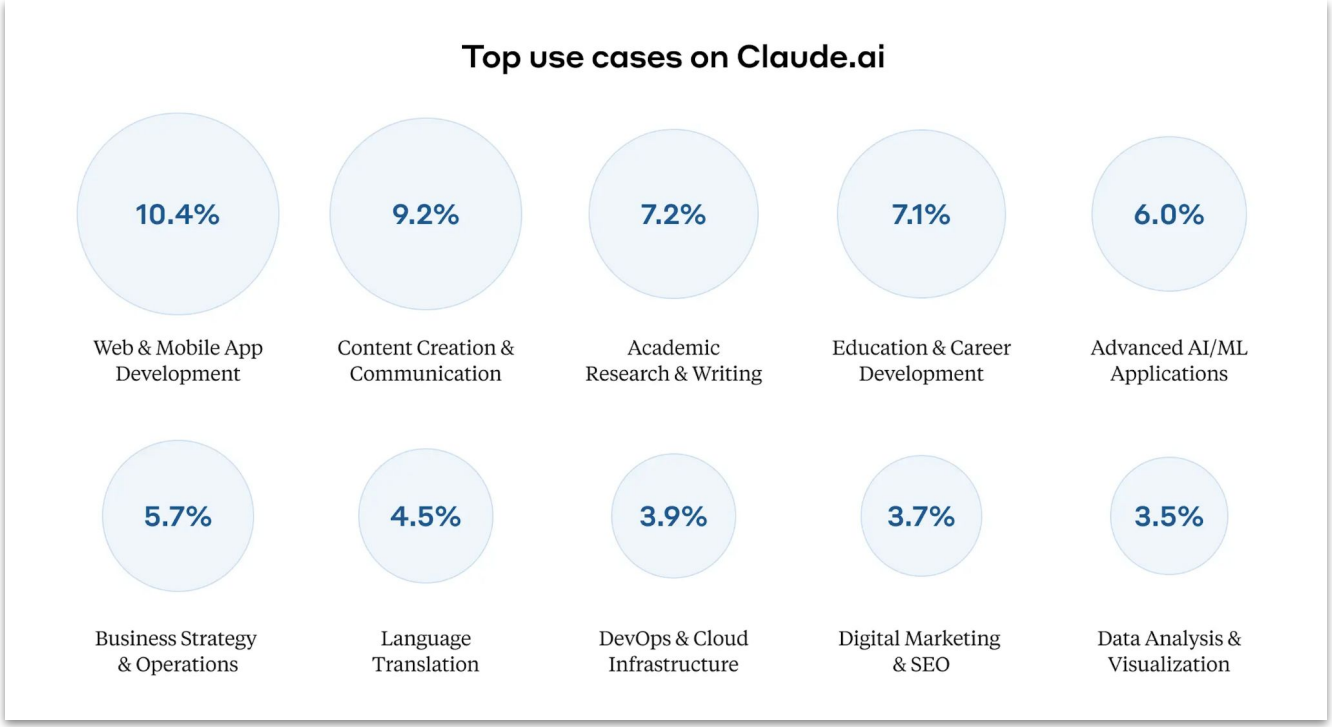
LLM Analytics

Clio (Claude Insights and Observations) Algorithm workflow



https://www.anthropic.com/research/clio

LLM Analytics



https://www.anthropic.com/research/clio

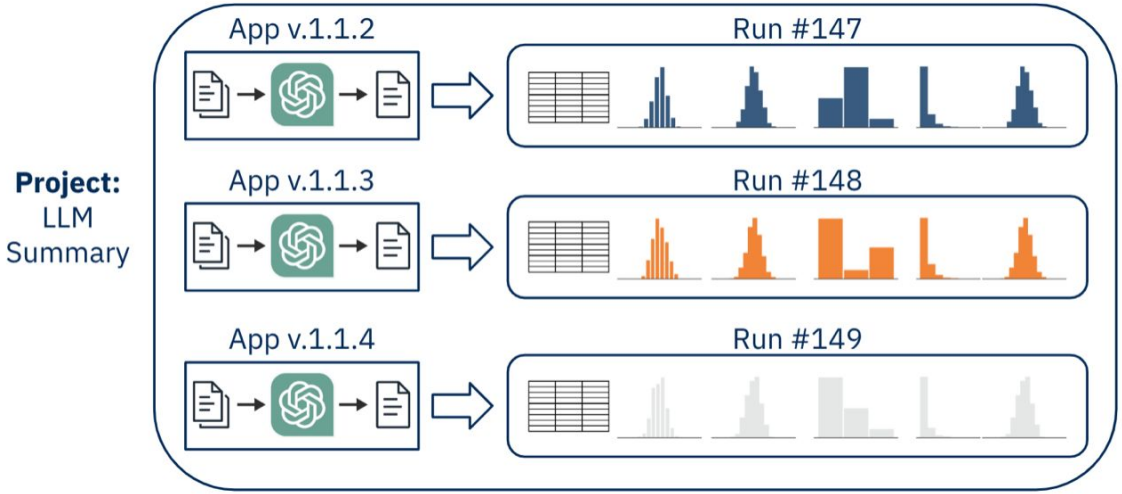
LLM Testing

Once we have a system for collecting data – whether that's from Evals or other Analytics, we can then observe that data over time and write tests on those metrics.

- With ML, it's sufficient to do offline evaluation in development and have some simple drift monitoring system in production.
- Due to new issues like silent model upgrades (no code change on your end, but the system changes), it is imperative that you "test in production".

LLM Eval: Distributional Testing in Production

Similar to how you need to test for data drift in the feature inputs and predictions of traditional ML systems, you also need to test input/output distributions over time with LLM systems in production.



- The output of LLMs is text based (structured outputs can contain other metadata)
- There are many derived metrics from LLM input/output which you can track over time.
- In contrast to ML, where tracking the distribution of a single predicted value is very straightforward.

06 Towards Auto-Eval

Can We Automate LLM Evaluation?

There are many strides towards this, however, **we are not currently in a place where you can do high quality LLM Evals in a fully automated fashion** for most applications (yet). Why not?

Automated metrics are limited

- Classic text metrics (ROUGE, BLEU) often fail to capture quality in open-ended tasks.

LLM-as-Judge still needs guidance

- Judges must be calibrated to domain goals and human preferences. They reduce cost, but don't eliminate human oversight.

Human judgment is still essential

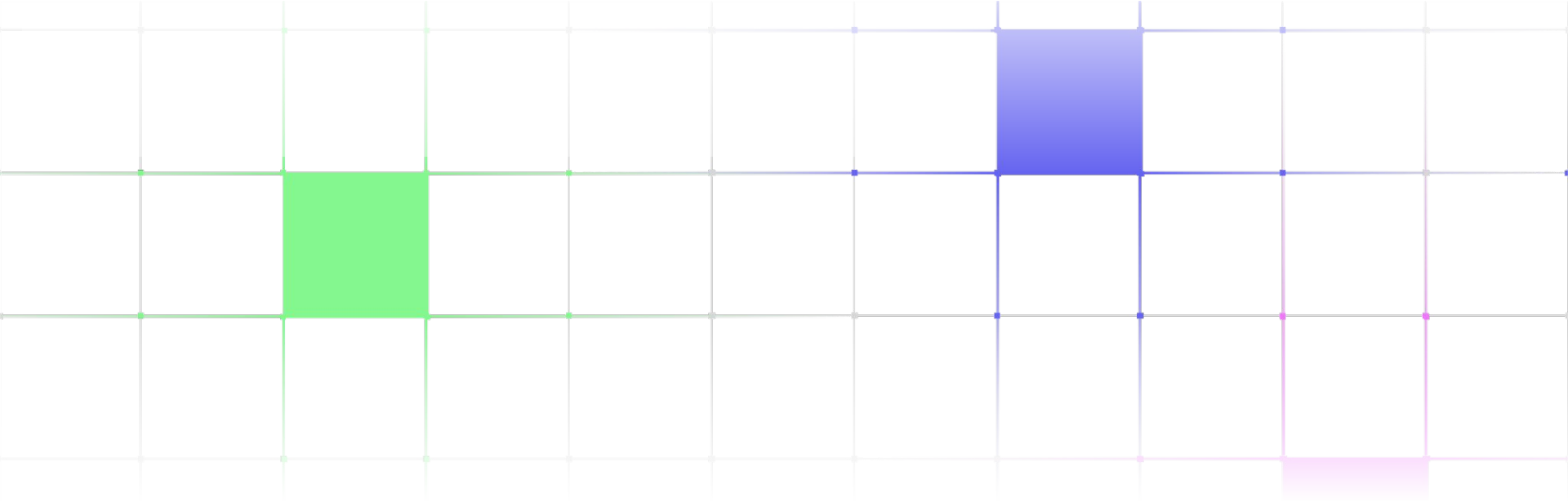
- Many dimensions of quality (safety, tone, usefulness) remain subjective. Current tools (e.g., EvalGen) make this efficient, but humans stay in the loop.

07 Resources

- Distributional: https://distributional.com
- Hamel Husain (LLM Eval): https://hamel.dev/
- Shreya Shankar (LLM Eval): https://www.sh-reya.com/

[DISTRIBUTIONAL]

Thank you!

Connect

https://www.linkedin.com/in/erin-ledell/